EXPERIMENTAL PAPER BLUEPRINT

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Programme & Institution:	MSc Data Science and Computational Intelligence | Softwarica College / Coventry University
Module & Assignment:	STW7072CEM / ST7072CEM Machine Learning Coursework CW (Regular)
Coursework Research Project:	**Machine Learning-Based Analysis of Daily Rainfall Patterns and Extreme Rainfall Events in the Kathmandu Valley**

Reference Paper Blueprint

- **Target Paper:** 'Spatial and temporal variation of precipitation extreme in the Kathmandu Valley during last three decades'
- **Citation:** Khanal et al. (2024)
- Paper link: https://doi.org/10.3126/jhm.v12i1.72653

How to use this worksheet

1. Check the average length of comparable papers on a similar topic or in the same journal look for papers close to yours in methodology.

Comparable paper	**Length (excl. references)**
Paper 1: Lei et al. (2025), Water 17(12), 1776	~8,400 words
Paper 2: Sharma et al. (2020), Earth and Space Science 7(8), e2020EA001315	~7,900 words
Paper 3: Khanal et al. (2024), J. Hydrol. Meteorol. 12(1), 34-45	~4,200 words
Paper 4	—
Paper 5	—
Average	**~6,800 words**

2. Adjust the overall suggested length using that average.
3. Adjust the length of each section based on the papers you reviewed.

4. Check which sections are used in your field and edit the table accordingly:
    - Which sections from the table are present? Delete any that aren't.
    - Is there a separate literature review section?
    - Are results and discussion one section, or two?
    - Are discussion and conclusion one section, or two?
5. Compare the elements listed in each section against papers in your field:
    - Are all the listed elements present? Delete any that aren't.
    - Are they presented in the same order? If not, reorder the table.
6. Save the completed table below as its own document and this becomes your personal blueprint for future empirical papers.

Blueprint

Suggested overall length:

- **Benchmark:** 4000-7000 words (without references)
- **Recommended range:** 4000-5500 words
- **Average length in my field:** ~4,200 words (based on JHM journal format)

Section	Element	Purpose	In my field?	My notes and ideas
Introduction(600-900 words)	Importance of the topic	Introduce the topic and outline why it should be studied	✓ Yes	**Precipitation** extremes trigger urban stormwater flooding, landslides, and infrastructure damage in the rapidly expanding Kathmandu Valley. Establishing **rainfall** thresholds and trends is crucial for municipal hazard mitigation.
	Brief literature review	Situate your study within the context of your field	✓ Yes	Reviews global and regional extreme **rainfall** trends (IPCC, ETCCDI indices); notes contradicting **precipitation** trends across Nepal and Koshi/Narayani river basins; highlights the flood history of the Bagmati River Basin.
	Research gap	Provide justification for your study	✓ Yes	Sparse multi-decadal studies evaluating specific ETCCDI extreme indices and dry/wet spell durations exclusively using in-situ weather stations within the Kathmandu Valley.

Section	Element	Purpose	In my field?	My notes and ideas
	Research aim	Present what your study was about	✓ Yes	Analyze annual and seasonal **precipitation** trends and 10 ETCCDI extreme **precipitation** indices across 5 meteorological stations in the Kathmandu Valley over a 33-year period (1990-2022).
	The main contributions of the paper (optional)	Highlight what novel insights your study offers	✓ Yes	Quantifies multi-decadal trends across the valley: identifies opposing trends (Godawari decreasing at -12.59 mm/yr vs Ktm Airport increasing at +2.63 mm/yr); documents sharp increase in consecutive dry days (CDD) across all stations.
	Structure of the paper (optional)	Present the organization of the paper	✓ Yes	JHM SOHAM format: Sec 2 describes Kathmandu Valley geography, **DHM** stations, MK test, Sen's slope, and ClimPACT2 indices; Sec 3 presents empirical trends; Sec 4 discusses thresholds and meteorological causes; Sec 5 concludes.
Literature Review(optional 1000-1500 words)	Key themes from the literature / Definitions of key concepts	Outline the literature on the main topics related to your research	✓ Yes	In regional hydrology journals like JHM, the literature review is concise and directly integrated within the Introduction and Discussion.
	Critical analysis	Analyze previous studies critically to reinforce the research gap	✓ Yes	Contrasts national-scale studies showing declining **precipitation** with station-level urban analyses showing localized increases in short-duration extreme events.
Theoretical / Methodological Framework(optional 200-500 words)	The theories utilized in the study	Present which theories you drew upon when designing the study and/or analyzing the data	✓ Yes	Non-parametric rank-based trend testing (Mann-Kendall test statistic S, variance, and Zc score); Sen's non-parametric slope estimator (Qmed); WMO CCI/WCRP ETCCDI **climate** extreme definitions.

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Methodology(800-1200 words)	Definition and justification of the research methodology	Outline and justify the chosen methodology	✓ Yes	Non-parametric statistical methods (Mann-Kendall and Sen's slope) are distribution-free and robust against outliers and missing data. ETCCDI indices via ClimPACT2 provide internationally standardized **climate** metrics.
	Background information about the context of your study (optional)	Present the context in which your study was conducted (e.g. geography)	✓ Yes	Kathmandu Valley (~665 km², average elevation 1350 m asl), Bagmati Province, central Nepal. Surrounded by Shivapuri, Nagarjun, Phulchowki, and Chandragiri hills; temperate **climate** dominated by the summer monsoon (70-80% annual rain).
	Sample and sampling techniques + ethical considerations (if human/animal subjects)	Outline who or what you studied, how you got it, and any ethical considerations	✓ Yes	5 **DHM** weather stations: Kathmandu Airport (1030), Panipokhari (1039), Khumaltar (1029), Godawari (1022), and Bhaktapur (1052). Daily **rainfall** records from 1990 to 2022 (33 years). Government observational data properly acknowledged.
	Research tools and procedures	Specify which research tools you used and specifically how you conducted your study	✓ Yes	R software (ClimPACT2 package v1.2.8), Excel, ArcGIS. Data quality control and **validation** (CSV format with standard ETCCDI columns); computation of 10 indices; Sen's slope calculation for trends.
	Data analysis techniques	Present how you analyzed your data and justify your choice	✓ Yes	Evaluates 10 indices categorized into High Intensity **Precipitation** (RX1day, RX5day, R95, R99), Frequency (R10, R20), Duration/Spells (CDD, CWD), and Extra (PRCPTOT, SDII). **Spatial** mapping of seasonal extremes via GIS interpolation.
Results(1200-1800 words)	Presentation of the main results	Present the results you obtained	✓ Yes	(1) Annual **rainfall**: Godawari (-12.59 mm/yr, p=0.023) and Bhaktapur (-4.57 mm/yr) decreased; Airport (+2.63 mm/yr), Khumaltar (+1.95

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				mm/yr), Panipokhari (+4.45 mm/yr) increased; (2) **Spatial** extremes: Godawari received highest annual **rainfall** (1668.9 mm) and highest RX1/RX5; Khumaltar lowest (1162.4 mm); (3) Consecutive Dry Days (CDD) increased across all 5 stations (0.55 to 1.29 days/yr); (4) Frequency of heavy days (R10) decreased in Godawari (-0.26 days/yr) but remained high in count (70 days).
	Clear use of tables, figures or quotes from participants	Provide a visual representation of the data and raw evidence	✓ Yes	Table 1: ETCCDI index definitions; Table 2: Sen's slope values of 10 indices across 5 stations; Fig 1: Study area map with station locations; Fig 2: MK trend & Sen's slope graphs (1990-2022); Fig 3: Seasonal maximum **precipitation** maps; Fig 4: RX1day trend plots; Fig 5: R10 index trend plots; Fig 6: Monthly average **rainfall** cycle.
Discussion(800-1200 words)	Restatement of the result to be discussed	Remind the reader of the result	✓ Yes	Emphasizes the contrasting **spatial** patterns inside the valley: foothill stations (Godawari) show decreasing totals but intense single-day peaks, while central urban stations show increasing totals and rising dry spell lengths (CDD).
	Comparison of the result with the literature	Situate the result within the literature context	✓ Yes	Compares findings with national studies (Sigdel & Ma 2016, Karki et al. 2017) noting increased **precipitation** extremes over the central Himalaya; aligns with Prajapati et al. (2021) showing declining rainy days in Kathmandu.
	Explanation of the findings and/or differences with literature	Explain the findings you obtained and possible differences with previous studies	✓ Yes	Orographic uplift against Phulchowki hill drives excessive rain in Godawari; leeward depression lowers rain at Khumaltar; rising CDD alongside rising RX1day in urban

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				cores suggests concentrated high-intensity convective **rainfall** separated by longer dry spells.
	Interpretation of the findings	Show what your findings mean and how they can be understood	✓ Yes	Changing **rainfall** characteristics directly exacerbate urban flash flooding and drainage overload in Kathmandu, demanding dynamic stormwater management thresholds.
Conclusion(400-600 words)	Restatement of the main aim and topic	Remind the reader of the aim and topic of your study	✓ Yes	Examined 33-year (1990-2022) trends in daily **precipitation** extremes and ETCCDI **climate** indices across 5 stations in the Kathmandu Valley.
	Recap of the key findings and contributions + a brief discussion	Restate the main results and discuss briefly	✓ Yes	Godawari and Bhaktapur experienced declining **rainfall**, whereas central valley stations increased. Consecutive dry days (CDD) increased across all stations, confirming erratic **rainfall** redistribution.
	Practical implications of the findings	Show what impact your findings can have for different stakeholders	✓ Yes	Findings provide critical local thresholds for municipal urban flood risk management, storm drainage redesign, and **climate** adaptation planning in the valley.
	Limitations	Acknowledge the limitations of your study	✓ Yes	Reliance on only 5 stations; lacks sub-daily/hourly convective storm resolution; missing stations in northern hill crests.
	Suggestions for future research	Suggest what future research should focus on	✓ Yes	Integrate dense automatic weather station (AWS) networks; evaluate sub-daily **rainfall** thresholds for urban hydrological runoff models.